Tunneling as Reconfiguration Across Sparse ED-Gradients

Allen Proxmire
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Abstract
Tunneling is not motion through a barrier and not the traversal of a forbidden region. In the ED ontology, a barrier is a region of sparse ED-gradients that cannot individuate the system. A low-multiplicity configuration cannot stabilize inside such a region, so the system's participation rule reconfigures globally between the nearest stable ED-structures. The system does not enter the barrier, spend time inside it, or cross it; the barrier interior is a topological gap in the ED-landscape, not a physical location the system occupies. The exponential suppression of tunneling probability reflects the structural incompatibility between the stable regions on either side of the sparse zone, not the spatial width of the barrier. Tunneling is fast because no traversal occurs, and macroscopic systems do not tunnel because their high-multiplicity ED-structure requires continuous support that sparse regions cannot provide. Measurement does not collapse tunneling; it individuates the outcome of a process whose structural conditions were already fixed by the ED-gradients. Tunneling is therefore not a quantum anomaly but the structural signature of sparse ED-structure: a system appears on the far side of a barrier not because it has crossed it, but because that is the nearest region capable of supporting its identity.

1. Standard Quantum Background
In the conventional quantum picture, tunneling is introduced as a surprising but mathematically inevitable consequence of wavefunction evolution. A particle with energy $E < V_{\text{barrier}}$ can still appear on the far side of a potential barrier, even though classical mechanics forbids it. The wavefunction decays exponentially inside the barrier but never reaches zero, so there is always a nonzero probability of transmission. The particle is said to "tunnel through" the barrier, even though the barrier is a region the particle cannot classically enter.

Quantum mechanics predicts tunneling with extraordinary accuracy, but it offers no physical mechanism. The wavefunction is treated as a mathematical object that evolves smoothly through regions where the particle cannot exist. The barrier interior is assigned amplitude even though the particle cannot occupy it. The event is continuous, unitary, and probabilistic, but the theory does not specify what physically happens inside the barrier or how the system "moves" through a region that cannot support it.

Several conceptual gaps follow:

- Existence without individuation.
    - The wavefunction assigns amplitude to regions where the system cannot form a stable configuration, but the theory does not explain what this amplitude represents physically.
- Traversal without mechanism.
    - The particle is said to cross the barrier, but no mechanism is given for how it moves through a region it cannot inhabit.
- Continuity without structure.
    - The wavefunction evolves smoothly even when the underlying physical structure cannot support the system's identity.
- Timing without presence.
    - Experiments that measure tunneling times reveal apparent "superluminal" features, but the theory cannot explain how a particle spends time in a region it cannot occupy.

Quantum mechanics treats tunneling as a mathematical consequence of the Schrödinger equation. It predicts the probabilities but does not explain what tunneling is. The result is a picture in which a system appears on the far side of a barrier without ever having a physical presence inside it, and in which the barrier interior is simultaneously forbidden and occupied. Tunneling is indispensable to the theory but conceptually opaque: a process that behaves like motion without motion, that assigns amplitude without structure, and that produces outcomes without a mechanism.

In the ED ontology, tunneling is not motion and not traversal. It is the global reconfiguration of a participation rule across a region that cannot individuate the system. The barrier interior is not a place the system occupies; it is a region where ED-gradients are too sparse to support identity. The system does not pass through the barrier. It persists across a gap in ED-structure.

2. ED Ontology: The Relevant Components

Tunneling in ED does not require new machinery. It emerges directly from the same structural elements that govern identity, individuation, and participation everywhere in the ontology. Only a few components are needed, and each plays a precise role in explaining why tunneling occurs and why it looks the way it does.

(1) ED-Gradients as the Structure of Existence

A system exists only where ED-gradients can individuate it. These gradients determine what the system can do, how it can evolve, and where it can stabilize. A region with rich ED-structure can support identity; a region with sparse ED-structure cannot. Existence is not a matter of occupying space but of being supported by ED-gradients.

(2) Sparse Regions as Non-Individuating Zones

A "barrier" is not a wall. It is a region where ED-gradients are too weak, too flat, or too incoherent to individuate the system. The system cannot complete its participation rule inside such a region. The barrier interior is therefore not a place the system can exist. It is a structural gap in the ED-landscape.

(3) Participation Rules as Identity Constraints

Every system carries a participation rule: the minimal structural constraint that determines how it engages with ED-gradients. This rule specifies where the system can stabilize and how it can persist. If a region cannot support the rule, the system cannot exist there. Tunneling occurs when the participation rule remains coherent across a region that cannot individuate it.

(4) Global Reconfiguration Instead of Local Motion

When a system encounters a region where its participation rule cannot be completed, it does not move through that region. It reconfigures globally between the nearest stable ED-structures. The system "appears" on the far side not because it has crossed the barrier but because its participation rule spans the gap between two regions that can support it.

(5) Exponential Suppression as Structural Incompatibility

The tunneling amplitude reflects how compatible the two stable ED-regions are with the system's participation rule. A large mismatch produces strong suppression; a small mismatch produces weak suppression. The exponential decay is not about spatial distance but about structural distance: how difficult it is for the participation rule to remain coherent across the sparse region.

(6) Saddles as orthogonal ED-gradients directions
The geometric form created when ED-gradients flip sign across orthogonal directions. Boundaries where compression dominates and horizons where expansion dominates. Sparse regions lack the gradients needed to form saddles, so they cannot stabilize identity.

These components are sufficient to reinterpret tunneling as a structural phenomenon. The discreteness, the suppression, the timing, and the apparent paradoxes all follow from the same architectural fact: the system cannot exist where ED cannot individuate it. Tunneling is therefore not a quantum oddity. It is the structural signature of sparse ED-gradients.

3. What a Barrier Is in ED

In the ED ontology, a barrier is not an obstacle the system must cross and not a region the system can partially occupy. A barrier is a region of sparse ED-gradients—a structural gap where the system's participation rule cannot be completed. The barrier interior is therefore not a place the system can exist. It is a zone where identity cannot stabilize.

3.1 No Interior Region

Quantum mechanics treats the barrier interior as a region the wavefunction occupies, even though the particle cannot. ED rejects this picture. A system exists only where ED-gradients can individuate it. A sparse region cannot support identity, so the system does not enter it, traverse it, or spend time inside it. The interior of the barrier is not a location the system visits. It is a structural absence.

3.2 Sparse Gradients Cannot Individuate

A low-multiplicity system requires a minimal amount of ED-structure to maintain its participation rule. Inside a barrier, the gradients are too weak or too flat to support that rule. The system cannot complete its identity there. This is why the classical picture forbids entry: classical objects require continuous ED-support. But even quantum systems cannot exist inside the barrier. They simply do not need continuous support to reconfigure across it.

3.3 Why Classical Barriers Block

A classical object possesses high multiplicity and rich ED-structure. It requires continuous ED-support to maintain its identity. A sparse region breaks that continuity. The object cannot reconfigure because its identity is too developed to span the gap. Classical barriers block because classical identity is rigid.

3.4 Why Quantum Systems Can Reconfigure

A low-multiplicity system has a flexible participation rule. It does not require continuous ED-support. When it encounters a sparse region, it does not attempt to enter it. Instead, its participation rule remains coherent across the gap, allowing the system to reappear in the next region capable of individuating it. The system does not cross the barrier. It persists across it.

4. Tunneling as Global Reconfiguration

Tunneling is not motion through a forbidden region. It is the global reconfiguration of a participation rule across a zone that cannot individuate the system. The system does not traverse the barrier because traversal requires existence inside the barrier, and existence requires ED-structure the barrier does not possess.

4.1 No Traversal

The standard language of “crossing” or “passing through” the barrier is misleading. The system never occupies the sparse region. There is no trajectory, no path, and no intermediate state inside the barrier. The system appears on the far side because its participation rule spans two stable ED-regions separated by a gap.

4.2 Continuity of the Participation Rule

A participation rule is not a spatial trajectory. It is a structural constraint on how the system engages with ED-gradients. If the rule can remain coherent across a sparse region, the system can reconfigure globally between the nearest stable ED-structures. The continuity is not geometric. It is structural. The system persists because the rule persists.

4.3 Why the System “Appears” on the Other Side

The ED-network supports a stable configuration on both sides of the barrier but not in between. When the system encounters the sparse region, it cannot individuate locally. Instead, it completes its participation rule in the next region capable of supporting it. The system “appears” on the far side because that is the nearest location where its identity can stabilize.

4.4 The Barrier as a Topological Gap

The barrier is not a physical obstacle. It is a topological gap in the ED-landscape—a region where the ED-structure required for identity is missing. The system does not cross the gap. It spans it. The reconfiguration is global because the failure of individuation is global.

5. Exponential Suppression as Structural Incompatibility

In quantum mechanics, the tunneling probability decays exponentially with barrier width and height. This decay is treated as a mathematical consequence of the Schrödinger equation: the wavefunction attenuates inside the barrier, and the transmission probability reflects the amplitude that survives the decay. The theory predicts the numbers but does not explain what the decay means physically. Why should a system’s ability to appear on the far side of a barrier depend exponentially on a region it cannot occupy?

In the ED ontology, the exponential suppression is not a property of motion and not a measure of how far the system must travel. It is a measure of structural incompatibility between the two stable ED-regions separated by the sparse zone. The suppression reflects how difficult it is for the system’s participation rule to remain coherent across a region that cannot individuate it.

5.1 The Standard Decay

Quantum mechanics describes the barrier interior as a region where the wavefunction decays exponentially. The decay constant depends on the difference between the barrier height and the system’s energy. The transmission probability is proportional to the square of the attenuated amplitude. The mathematics is clear, but the physical meaning is not.

5.2 ED Interpretation: Compatibility of Participation Rules

A participation rule is a structural constraint. It specifies how the system can stabilize in the ED-landscape. When two stable ED-regions are separated by a sparse zone, the system can reconfigure between them only if the participation rule can remain coherent across the gap. The degree of coherence depends on how similar the two stable regions are in ED-terms. If the regions are structurally compatible, the rule can span the gap easily. If they are incompatible, the rule struggles to persist.

The exponential suppression is the structural signature of this difficulty.

5.3 Structural Distance, Not Spatial Distance

The suppression does not measure how far the system must travel. It measures how far apart the two stable ED-regions are in structural space. A wide barrier in physical space may correspond to a small structural mismatch if the ED-gradients on both sides are similar. A narrow barrier may correspond to a large mismatch if the ED-gradients differ sharply.

The exponential decay is therefore not geometric. It is architectural.

5.4 Why Some Barriers Are "Thin"

A barrier is "thin" when the two stable ED-regions are structurally similar. The participation rule can span the gap with little suppression. A barrier is "thick" when the regions are structurally dissimilar. The rule struggles to remain coherent, and the suppression is strong.

6. Why Tunneling Is Fast

Tunneling appears to occur with little or no time spent inside the barrier. Experiments that attempt to measure tunneling times often report "superluminal" features or durations that do not scale with barrier width. These results are puzzling in the standard picture, where the particle is assumed to traverse the barrier interior. If the particle must cross a region it cannot occupy, how can it do so faster than light?

In the ED ontology, the puzzle dissolves. Tunneling is fast because there is no traversal. The system does not move through the barrier. It reconfigures globally between the nearest stable ED-regions. The sparse region plays no dynamical role because the system never exists inside it.

6.1 No Time Spent Inside the Barrier

Time is defined only where the system exists. A system cannot spend time in a region where it cannot individuate. The barrier interior is a structural gap, not a location. Since the system never occupies the sparse region, it cannot accumulate temporal duration there. The question "How long does the system spend in the barrier?" has no physical meaning in ED.

6.2 Reconfiguration Is Instantaneous Relative to Local Motion

Local motion requires continuous ED-support. Reconfiguration does not. When the system encounters a sparse region, its participation rule fails locally but remains coherent globally. The system completes its identity in the next region capable of supporting it. This reconfiguration is not instantaneous in an absolute sense, but it is instantaneous relative to any process that requires local propagation.

The system does not move. It resolves.

6.3 Why Experiments See "Superluminal" Features

Experiments that measure tunneling times often interpret the event as motion. They assume the system must cross the barrier and therefore attempt to assign a traversal time. But the system does not traverse the barrier. It reconfigures across it. The measured times reflect the dynamics of reconfiguration, not the dynamics of motion. Because reconfiguration does not require propagation through the sparse region, the inferred traversal time can appear shorter than the time light would require to cross the same spatial distance.

There is no violation of relativity. There is only a misinterpretation of what the system is doing.

7. Macroscopic Tunneling

Macroscopic tunneling is often described as a rare or exotic extension of quantum behavior into the classical world. In the standard picture, tunneling is a universal quantum process that becomes negligible for large systems because the wavefunction decays too quickly inside the barrier. The mathematics predicts the suppression, but the theory does not explain why macroscopic systems should be fundamentally different from microscopic ones. Why should size matter if the underlying rules are the same?

In the ED ontology, the distinction is structural, not statistical. Macroscopic systems possess high multiplicity and rich ED-gradients. Their identity is fully developed, and their participation rules require continuous ED-support. Sparse regions break that continuity. A macroscopic system cannot reconfigure across a gap because its identity is too rigid to span it. Tunneling is therefore not suppressed because the system is large. It is suppressed because the system is structurally complex.

7.1 Why Macroscopic Systems Rarely Tunnel

A macroscopic system is a high-multiplicity ED-object. Its identity is encoded in a dense network of ED-gradients that must remain coherent for the system to persist. A sparse region cannot support this network. The system cannot maintain its participation rule across the gap, so tunneling is structurally impossible. The system is not "unlikely" to tunnel. It is incapable of tunneling because the ED-structure required to span the gap does not exist.

(Saddle note: high-multiplicity systems require continuous saddle-like gradient transitions to maintain identity. Sparse regions cannot form these transitions, so macroscopic identity cannot propagate across them.)

7.2 When Macroscopic Tunneling Occurs

Macroscopic tunneling occurs only when a macroscopic system behaves, in ED-terms, like a low-multiplicity one. This happens when the system's identity is encoded in a collective participation rule rather than in the detailed structure of its constituents. Superconductors, superfluids, and Josephson junctions exhibit tunneling because their collective ED-structure is simple enough to span sparse regions. The system tunnels not because it is macroscopic, but because its effective participation rule is low-multiplicity.

7.3 ED Interpretation of Josephson Tunneling

In a Josephson junction, the superconducting phase is a collective participation rule that spans the junction. The insulating barrier is a sparse region that cannot individuate the superconducting identity, but the phase coherence can remain intact across the gap. The system does not send particles through the barrier. It maintains a global participation rule that spans two stable ED-regions separated by a sparse zone. The Josephson current is the expression of this rule.

Macroscopic tunneling is therefore not a quantum anomaly. It is the structural signature of a system whose effective identity is simple enough to reconfigure across a sparse ED-region.

8. Tunneling and Measurement

In the standard picture, measurement plays no special role in tunneling. The wavefunction evolves unitarily, and the measurement simply reveals whether the particle has appeared on the far side of the barrier. The theory does not explain why tunneling probabilities are stable, why measurement does not disrupt the process, or why the barrier interior remains unoccupied even though the wavefunction assigns amplitude to it.

In the ED ontology, measurement is an ED-proliferation event. A measurement device injects ED-structure into the system, forcing individuation. But tunneling probabilities are determined by the structural compatibility of the stable ED-regions, not by the act of measurement. Measurement does not collapse tunneling. It simply individuates the outcome of a process whose structural conditions were already fixed.

8.1 Measurement as ED-Proliferation

A measurement device is a high-multiplicity ED-object. When a low-multiplicity system interacts with it, ED-flow proliferates, forcing the system to individuate relative to the device's gradients. This individuation determines the outcome but does not alter the structural compatibility of the regions involved in tunneling. The measurement does not change the tunneling probability because the probability is not a property of the system's state. It is a property of the ED-landscape.

8.2 Why Tunneling Probabilities Are Stable

The tunneling probability is determined by the structural relationship between the two stable ED-regions separated by the sparse zone. This relationship does not depend on measurement. It depends on the ED-gradients that define the barrier and the regions on either side. Measurement reveals the outcome but does not influence the structural compatibility that determines the probability.

8.3 Why Measurement Does Not "Collapse" Tunneling

In the ED ontology, there is no wavefunction to collapse. There is only the system's participation rule and the ED-structure that supports it. The system reconfigures across the sparse region if and only if the participation rule can remain coherent. Measurement does not alter this coherence. It only individuates the system after the reconfiguration has occurred. The outcome is not created by measurement. It is completed by it.

9. Why Tunneling Has No Classical Analogue

In classical physics, tunneling is impossible. A classical object cannot appear on the far side of a barrier unless it has enough energy to cross it. The impossibility is not a matter of probability. It is a matter of structure: classical identity is too rigid to span a region that cannot support it. The standard quantum picture treats tunneling as a uniquely quantum effect, a phenomenon that emerges only when the wavefunction is allowed to extend into forbidden regions. But this explanation does not reveal why classical objects lack the ability to tunnel or why quantum systems possess it.

In the ED ontology, the distinction is structural and immediate. Classical objects possess high multiplicity and dense ED-gradients. Their identity is encoded in a continuous, richly developed ED-structure that requires uninterrupted support. A sparse region breaks that support. The object cannot maintain its participation rule across the gap, so tunneling is structurally impossible. Quantum systems, by contrast, are low-multiplicity. Their identity is simple enough to reconfigure globally when local stabilization fails. Tunneling is therefore not a quantum anomaly. It is the behavior of systems whose identity is not yet fully developed.

9.1 Classical Objects Require Continuous ED-Support

A classical object is a high-multiplicity ED-entity. Its identity is encoded in a dense network of ED-gradients that must remain coherent for the object to persist. Every small change in position corresponds to a small change in ED-structure. The object can move only through regions that can support this structure. A sparse region cannot. The object cannot enter it, cannot span it, and cannot reconfigure across it. Classical motion requires continuous ED-support, and sparse regions break that continuity.

9.2 Low-Multiplicity Systems Can Reconfigure

A low-multiplicity system has a simple participation rule. It does not require continuous ED-support. When it encounters a sparse region, it does not attempt to enter it. Instead, its participation rule remains coherent across the gap, allowing the system to reappear in the next region capable of individuating it. The system does not cross the barrier. It spans it. This ability is not quantum. It is structural. It arises because the system's identity is simple enough to survive the absence of local ED-support.

9.3 Classical Intuition Fails Because It Assumes Multiplicity

Classical intuition assumes that identity is rigid, continuous, and geometrically localized. This assumption is correct for classical objects but incorrect for low-multiplicity systems. Tunneling appears paradoxical only when classical intuition is applied to systems that do not possess classical structure. The paradox dissolves when identity is understood as an ED-phenomenon. Classical objects cannot tunnel because their identity is too developed. Quantum systems can tunnel because their identity is not.

Tunneling has no classical analogue because classical identity has no structural flexibility. The phenomenon is not mysterious. It is the natural behavior of systems whose ED-structure is too simple to require continuous support.

10. Summary

In the ED ontology, tunneling is not motion through a barrier, not traversal of a forbidden region, and not a uniquely quantum effect. It is the global reconfiguration of a participation rule across a region that cannot individuate the system. A barrier is a sparse ED-region—a structural gap where identity cannot stabilize. The system does not enter the barrier, spend time inside it, or cross it. It persists across it because its participation rule remains coherent between the nearest stable ED-structures.

The exponential suppression of tunneling probability reflects structural incompatibility, not spatial width. The suppression measures how difficult it is for the participation rule to span the gap, not how far the system must travel. Tunneling is fast because there is no traversal. The system never occupies the sparse region, so it cannot accumulate temporal duration there. Apparent "superluminal" features arise from misinterpreting reconfiguration as motion.

Macroscopic systems do not tunnel because their identity is too developed to span sparse regions. Only systems with low-multiplicity ED-structure possess the flexibility required for global reconfiguration. Measurement does not collapse tunneling. It individuates the outcome of a process whose structural conditions were already fixed by the ED-landscape.

Tunneling is therefore not a quantum mystery. It is the structural signature of sparse ED-gradients. A system appears on the far side of a barrier not because it has crossed it, but because the ED-structure on that side is the nearest region capable of supporting its identity.

References

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